Cardinality

Are there more rational numbers than natural numbers? The rationals are dense, with a third between any two, while the naturals are spread out one unit apart; and yet the answer is no, because the rationals can be listed. Are there more real numbers than rationals? Yes: no list of real numbers, however cleverly it is made, contains them all. This chapter makes "more" exact, with bijections and injections in place of counting, and proves both answers. It then shows that there is no largest size at all: by Cantor's theorem, the set of subsets of any set is strictly bigger than the set itself. A second motivating question comes from school: *why does* $0.999\ldots = 1$ *need a definition at all?* The answer is in Section 4.3, where a decimal expansion is defined with finite sums only, and $0.999\ldots = 1$ becomes true by that definition. The chapter starts from finite sets: Section 4.1 defines "same size" by bijections, proves the pigeonhole principle and finishes the lattice-point puzzle of Section 1.1. Sections 4.1–4.4 teach one method of proof each: pigeonhole, list a set, run a diagonal, and compare sizes by injections both ways. The last section adds, multiplies and raises sizes to powers, absorbs a countable set into any set that contains a copy of $\mathbb{N}$, and finds that the plane and the subsets of $\mathbb{N}$ have exactly as many elements as the line. It teaches two more methods: encode and decode, and prove a law of sizes by a bijection.

Every map in this chapter is given by a rule: a formula, a least element, a recursion with an explicit step. Some natural arguments would have to pick one element from each of infinitely many sets, with no rule to say which: a right inverse for every surjection, a copy of $\mathbb{N}$ inside every infinite set, one listing for each of countably many countable sets. Where that happens, the chapter says so and leaves the statement open. Chapter 6 adds the principle that settles these statements, the axiom of choice.

What you need. *Logic and Proof*, Chapters 1–8, and *Sets and Functions*, Chapters 1–3. Induction (Logic and Proof, Axiom 4.2.1) from *Logic and Proof*, Chapter 4. From *Logic and Proof*, Chapter 5, the rule ∃E (Logic and Proof, Definition 5.6.1), which names one object at a time. From *Logic and Proof*, Chapters 2 and 6, excluded middle (Logic and Proof, Proposition 2.5.3) and the negation of quantifiers (Logic and Proof, Proposition 6.2.1). From *Logic and Proof*, Chapter 8, the catalogue of what a proof may not do (Logic and Proof, Example (What is not allowed)). The irrationality of $\sqrt{2}$ (Logic and Proof, Theorem 2.2.1), used in Sections 4.2 and 4.5. From Chapters 1 and 2: the empty set and its uniqueness (Definition 1.2.4, Proposition 1.2.3); sets cut out by a property (Definition 1.2.3) and Russell's theorem (Theorem 1.2.4); power sets, families and their unions (Definition 1.3.2, Definition 1.3.3, Definition 1.3.4); ordered pairs and products (Definition 1.4.1, Definition 1.4.2, Proposition 1.4.1); functions, the empty function and equality of functions (Definition 2.1.1, Proposition 2.1.2, Proposition 2.1.1); injective, surjective and bijective maps and their composites (Definition 2.2.1, Definition 2.2.2, Theorem 2.2.1), with the methods for proving injectivity and surjectivity (Example (Method: proving injectivity), Example (Method: proving surjectivity)); inverses (Definition 2.4.1, Theorem 2.4.4, Theorem 2.4.1, Proposition 2.4.2); images and preimages (Definition 2.5.1, Definition 2.5.2); indicator functions and $|\mathcal{P}(X)| = \left|\{0,1\}^X\right|$ (Definition 2.3.2, Theorem 2.3.5); restriction, extension and gluing, and injectivity across the pieces (Definition 2.3.1, Theorem 2.3.2, Proposition 2.3.3); and associativity of composition (Proposition 2.1.3). From Chapter 3: the arithmetic of $\mathbb{N}$ and $\mathbb{Z}$ (Lemma 3.1.1, Proposition 3.1.2), well-ordering (Theorem 3.1.3) and the method of the least element (Example (Method: the least element)), induction from any base (Theorem 3.2.1), the division algorithm (Theorem 3.2.3), definition by recursion (Theorem 3.3.1), powers and sums (Definition 3.3.2, Proposition 3.3.4, Definition 3.3.1, Proposition 3.3.6), and the largest element of a finite set of numbers (Proposition 3.3.8). The arithmetic and order of $\mathbb{R}$ are school facts in examples, as in the earlier chapters. Section 4.3 states three school facts about decimal expansions, (D1)–(D3), and Sections 4.3, 4.4 and 4.5 use them by name without proof.

Time. About 30 hours on the core track (all of A, one B and one C per section, the exit checks); about 45 hours for everything.

4.-1.1 Roadmap

- **Finite sets.** Equinumerous sets, finite and infinite sets, the pigeonhole principle and the lattice points of Section 1.1, subsets and disjoint unions of finite sets, 2^n subsets, and no finite set the size of a proper subset; the method: pigeonhole.
- **Countable sets.** Countable as listable with repetitions allowed (empty, or the image of $\mathbb{N}$), the equivalent forms "an injection into $\mathbb{N}$" and "finite or countably infinite", closure under subsets, images, unions of two sets and finite products, and the countability of $\mathbb{N}\times\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$; the method: list a set, by a surjection from $\mathbb{N}$ or an injection into $\mathbb{N}$.
- **Diagonal arguments: uncountable sets and Cantor's theorem.** The diagonal argument for 0–1 sequences, opening with the five-string game of Section 2.1, Cantor's theorem that no map from a set onto its power set exists, decimal expansions defined with finite sums, and the uncountability of $\mathbb{R}$; the method: the diagonal.
- **Comparing sizes: the Schröder–Bernstein theorem.** Comparing sizes by injections, an injection gives a surjection back, the Schröder–Bernstein theorem proved without choice, and every interval with more than one point equinumerous with $\mathbb{R}$; the method: injections both ways; whether any two sets are comparable is left open.
- **Infinite sets and cardinal arithmetic.** A copy of $\mathbb{N}$ absorbs any countable set, with no choice made; sums as disjoint unions, products and powers compared by size; the laws $\left|X^{Y\sqcup Z}\right| = \left|X^Y \times X^Z\right|$ and $\left|(X^Y)^Z\right| = \left|X^{Y\times Z}\right|$ proved by bijections; no set is the size of its 0–1 functions; no set of all sets, once more; $\mathbb{R}$, $\mathbb{R}\times\mathbb{R}$, $\{0,1\}^{\mathbb{N}}$ and $\mathcal{P}(\mathbb{N})$ all of one size; the methods: encode and decode, and prove a law of sizes by a bijection. The review exercises reach back across this volume and *Logic and Proof*.

4.-1.2 Named moves

Each is a method box in the section named at the end of its line; the later sections use the move again.

- **Pigeonhole** (Example (Method: the pigeonhole principle)). Choose pigeons and fewer holes, define "the hole of a pigeon", and conclude that two pigeons share a hole. (Section 4.1)
- **List it** (Example (Method: list it)). To show a set countable, give a surjection from $\mathbb{N}$ or an injection into $\mathbb{N}$, whichever is cheaper to check; a list may repeat, and an injection only has to keep different elements apart. (Section 4.2)
- **The diagonal** (Example (Method: the diagonal)). Given any list, build an object that differs from the n-th entry at the n-th place; it is not on the list, and since the list was arbitrary, no list contains everything. (Section 4.3)
- **Injections both ways** (Example (Method: injections both ways)). To show $|A| = |B|$, find an injection $A \to B$ and an injection $B \to A$ and invoke the Schröder–Bernstein theorem, instead of hunting for the bijection. (Section 4.4)
- **Encode and decode** (Example (Method: build a bijection by encoding and decoding)). To build a bijection, say what an element must record, write the code and the decoder, and check both round trips. (Section 4.5)
- **A law by a bijection** (Example (Method: prove a law of sizes by a bijection of sets)). To prove a law of sizes, write out an element of each side and regroup it into an element of the other side, never computing with $|\cdot|$ as if it were a number. (Section 4.5)

4.-1.3 Threads

- **Lattice points.** Section 1.1 left a puzzle about five lattice points half solved; Section 4.1 finishes it with the pigeonhole principle (Example (Back to the five lattice points)).
- **The five strings.** Section 2.1 plants a game: write a string of five symbols not on a given list of five. Section 4.3 opens with the rule that always works (Example (Back to the five strings)), and the same rule, run on infinitely long strings, proves that the 0–1 sequences cannot be listed (Theorem 4.3.1).
- **Russell to Cantor.** Russell's theorem (Section 1.2) built a set that disagrees with each candidate on the candidate itself; Cantor's theorem (Section 4.3, Theorem 4.3.2) is the same construction, as a theorem about sizes, and Section 4.5 proves the last clause of Russell's theorem by sizes: a set of all sets would carry a surjection onto its power set ("No set of all sets, once more").
- **Choice.** Section 2.4 could not build a right inverse of an arbitrary surjection. Section 4.2 avoids choice with least preimages, Section 4.4 turns an injection into a surjection and leaves the converse open, and Section 4.5 absorbs countable sets only into sets that come with a copy of $\mathbb{N}$. Chapter 6 adds the axiom of choice and settles all three.
- **Comparability.** Section 4.4 asks whether any two sets are comparable and leaves the question open; Chapter 6 answers it.
- **Same size.** The bijections of Section 2.4 say when two sets have the same size; Section 4.1 makes this a definition, Section 4.5 makes it arithmetic, and Section 5.2 names the requirement that arithmetic of sizes not depend on representatives: well-definedness.

4.-1.4 Summary

Two sets have the same size when there is a bijection between them (Definition 4.1.1). A finite set has one number of elements (Corollary 4.1.4), and there is no injection from a finite set into one with fewer elements (Corollary 4.1.3). Countable means listable (Definition 4.2.1), in three equivalent forms (Theorem 4.2.2). Countable sets are closed under subsets, images, unions of two sets and finite products, so $\mathbb{Z}$ and $\mathbb{Q}$ are countable (Theorem 4.2.8). The diagonal shows that the 0–1 sequences, $\mathcal{P}(\mathbb{N})$ and $\mathbb{R}$ are too big to be listed, and that no set maps onto its power set (Theorem 4.3.1, Corollary 4.3.3, Theorem 4.3.4, Theorem 4.3.2). Sizes are compared by injections. An injection from a non-empty set gives a surjection back (Proposition 4.4.2), and injections both ways give a bijection (Theorem 4.4.3). A copy of $\mathbb{N}$ absorbs any countable set (Proposition 4.5.1). Sizes add, multiply and exponentiate, with the laws of exponents proved by bijections (Theorem 4.5.4); so $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$, and the plane is the size of the line (Theorem 4.5.6, Theorem 4.5.7). No proof in the chapter uses the axiom of choice.

What is open: does every infinite set contain a copy of $\mathbb{N}$? Does a surjection give an injection back? Are any two sets comparable? Chapter 6 answers all three with the axiom of choice. Next, Chapter 5 studies relations, and "same size" returns there as an example.

4.-1.5 Terms

- **at most as big as, strictly smaller**: $|A| \leq |B|$ if there is an injection $A \to B$; $|A| < |B|$ if moreover there is no bijection (Definition 4.4.1).
- **countable**: empty, or the image of a surjection $\mathbb{N} \to A$ (Definition 4.2.1).
- **countably infinite**: equinumerous with $\mathbb{N}$ (Definition 4.2.2).
- **decimal expansion** of $x \in [0,1]$: a digit sequence with $t_n \leq x \leq t_n + 10^{-n}$ for every $n \geq 1$, where t_n is the finite sum of the first n digits (Definition 4.3.1).
- **digit sequence**: a function from $\mathbb{Z}_{>0}$ to $\{0, 1, \ldots, 9\}$ (Definition 4.3.1).
- **disjoint union** $A \sqcup B$: the set $(A \times \{0\}) \cup (B \times \{1\})$ (Definition 4.5.1).

- **equinumerous**: there is a bijection $A \to B$, written $|A| = |B|$ (Definition 4.1.1).
- **finite**: equinumerous with $\{1, \ldots, n\}$ for some $n \in \mathbb{N}$ (Definition 4.1.2).
- **infinite**: not finite (Definition 4.1.3).
- **uncountable**: not countable (Definition 4.2.1).